

Project Design Proposal: Assistive Backpack for Independent Living

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ABSTRACT

This project proposes an assistive backpack designed to support independent completion of daily activities for people experiencing memory and associated cognitive difficulties. The system combines RFID item detection system with a simple user interface to help users identify and pack the belongings required for selected tasks. An Arduino Uno acts as the central controller, using a finite state machine to manage task selection, item detection and system feedback. A 16×2 LCD provides task and checklist information, while an RGB LED and buzzer provide additional visual and audible feedback when items are detected or missing. The system is designed to operate from a rechargeable battery for at least eight hours, with RFID detection targeted at 95% accuracy. Development will involve independent subsystem testing followed by system integration and validation against defined engineering specifications. The proposed design prioritises simplicity, portability, reliability and low power consumption while providing assistance without unnecessarily reducing user independence.

1 Project Overview

Maintaining independence during everyday activities can become increasingly difficult for those suffering with cognitive or mental health challenges such as dementia. Tasks such as remembering personal belongings, following a daily routine, and preparing to leave the house can become barriers to participating in social and communal activities, as well as ones independence. This project proposes a wearable backpack assistive system designed to provide simple and intuitive reminders while helping users organise tasks for their day. By combining RFID identification, visual feedback, buttons, and audible alerts, the system aims to support users in completing daily routines independently while providing peace of mind for carers.

1.1 Problem Statement

People living with dementia or other conditions that affect memory, motivation, and routine may have difficulty completing daily tasks independently. In Australia, approximately 63% of people with dementia were living in the community in 2022 [1], demonstrating the importance in assistive technology to support the large amount of people living outside of residential care. Forgetting essential items before leaving the house, missing planned activities or becoming unsure of what needs to be completed can reduce both confidence and independence. There is therefore an opportunity for a simple and portable system that assists users with remembering required belongings and daily tasks without requiring continuous assistance and supervision from another person.

1.2 Proposed Solution

The proposed solution is an intelligent backpack which incorporates an RFID based inventory tracking system, LCD display, physical buttons, and audible alerts. Users will be able to select a predefined daily task or a checklist using the backpack's buttons with each of the required items appearing on the display as a worded and visual aid. RFID tags attached to important belongings will allow the backpack to automatically detect whether required items have been placed inside, with the checklist providing visual feedback to reflect this. If the user attempts to leave without a required item, the system can provide an

audible warning. The system will therefore combine task reminders with automatic item verification to reduce the cognitive load associated with preparing for daily activities while maintaining a simple and intuitive interface.

1.3 Project Scope

The project will focus on developing a functional prototype demonstrating the core reminder and inventory management functions of the backpack.

In Scope

- RFID detection of tagged items placed inside the backpack
- An LCD display for visual task and inventory information
- Red and green visual feedback indicating whether required items have been identified
- Physical buttons for selecting between daily tasks or checklists
- Storage of at least three different tasks using EEPROM
- Audible alerts when required items are missing or another warning condition is detected
- Battery powered operation
- 8-12 hour operating period
- Intuitive user interface using simple visual cues that can be understood with minimal or no instruction

Stretch Goals

- Smartphone application functionality
- Scheduled tasks and reminders
- Carer monitoring and notifications
- Light sensor to provide an evening reminder or indication that it is becoming late
- Storage and display of medical identification or emergency information
- Disturbance sensor to make sure / warn that user has all required items before a trip

Out of Scope

- Providing navigation or way finding to locate individual items
- Fully autonomous location tracking of the user
- A wireless charging system using a charging plate integrated into the user's home environment

2 Background

Dementia represents a significant and growing health challenge to Australians every day. In 2024, dementia was Australia's leading cause of death, accounting for 17,550 registered deaths and 9.4% of all deaths [2]. Although dementia can significantly limit memory and independence, many people living with dementia continue to live in the community rather than in residential care. The Australian Institute of Health and Welfare reported that almost two-thirds of people diagnosed with dementia were living in the community in 2022 [1]. This creates a need for technologies that can provide assistance during everyday activities while allowing users to remain involved in their communities.

Maintaining social connection and mental stimulation is also an important requirement when designing support for people experiencing cognitive decline. Dementia Australia identifies social connection, mental stimulation, and maintaining an active lifestyle as factors associated with maintaining brain health and reducing dementia risk [3]. The proposed backpack is therefore intended not as a medical treatment or preventative device, but as an assistive technology that can encourage users to continue participating in daily activities and social routines. By helping a user prepare for an activity independently, the system may reduce one of the practical barriers to leaving the house and maintaining a regular routine.

The design is based on the principle of providing assistance without unnecessarily removing independence. Rather than requiring a carer to manually check whether the user has everything they need, RFID technology can automatically verify the presence of selected belongings. The user can then receive a simple visual indication of whether the checklist is complete. This approach combines cognitive prompting with physical interaction, allowing the user to actively complete the preparation process rather than simply receiving an external reminder.

Existing assistive products demonstrate the value of reminders, simplified interfaces and monitoring technologies for people living with dementia. For example, the Tochie scheduled reminder speaker provides scheduled spoken messages for activities such as medication, personal care and household tasks, while allowing confirmations to be monitored through an associated application [4]. The Robin Day Clock provides customisable visual displays, scheduled reminders and audible announcements, with tactile buttons intended to make interaction easier for people with cognitive impairment [5]. Wearable tracking solutions are also available, such as the Smart Tracker Wrist and Ankle Band sold by Dementia Shop Australia, which uses an Apple Find My-compatible tracking tag to provide location information to carers [6].

However, these products generally address individual aspects of the problem, such as reminders, orientation or location tracking. The proposed backpack combines several of these concepts into a single portable system, with the additional capability of automatically checking whether specific physical belongings have been packed. This cre-

ates an opportunity to develop a less complex and cheaper prototype that focuses specifically on the transition between preparing for an activity and leaving the home.

The system will therefore investigate whether RFID based inventory detection can be effectively combined with a simple user interface to support everyday preparation. The design will prioritise clear visual feedback, minimal interaction requirements, portability and sufficient battery life. These are particularly important for the intended user group, as an overly complex interface could increase rather than reduce the cognitive burden of using the device.

3 Requirements and Specifications

3.1 Requirements

The requirements were developed from the primary design objective of assisting users with memory and routine-related difficulties while maintaining a simple and independent interaction. Functional requirements describe what the user should be able to do, while engineering specifications define measurable targets that can be tested during prototype validation.

3.2 Functional Requirements

The proposed backpack shall:

1. Allow the user to select at least three predefined daily tasks using physical buttons.
2. Display the selected task and the belongings required for that task on the LCD.
3. Detect RFID-tagged belongings placed inside the backpack.
4. Indicate whether each required item has been detected using clear visual feedback.
5. Provide a clear indication when all the required items have been detected for a selected task.
6. Provide an audible warning when the user attempts to complete a task while one or more required items are missing.
7. Retain the predefined task configurations when the backpack is powered off.
8. Operate from a rechargeable battery without requiring a continuous connection to the mains power.
9. Provide clear instructions so that a first-time user can understand the basic task-selection and item-checking process without detailed instructions.

3.3 Engineering Specifications

The functional requirements were translated into measurable engineering targets. The baseline specifications summarised in Table 1. These values are intended to provide objective pass/fail criteria for prototype testing.

The RFID accuracy target will be tested using repeated trials for each tagged item. Battery performance will be assessed by measuring the operating current and recording the time taken for the battery voltage to reach a selected minimum operating voltage. The specifications avoid requiring continuous user location tracking / automatic detection of the user's departure from the home, as these functions are outside the core prototype scope.

Table 1: Engineering specifications and verification targets.

Parameter	Target	Priority
Task storage	≥3 task configurations retained after power removal	Baseline
RFID detection	≥95% successful detection over 20 trials per tagged item	Baseline
RFID response	Checklist updates within 2 seconds of tag detection	Baseline
Task selection	Any stored task selectable within 4 button presses	Baseline
Button response	System response within 300 ms of valid button press	Baseline
Checklist feed-back	Distinct complete/incomplete visual indication	Baseline
Missing-item warning	Audible warning within 2 s when a required item is missing	Baseline
Battery runtime	≥8 h continuous operation from a full charge	Baseline
Target runtime	10 h nominal operating target	Desirable
Power consumption	Average current measured during representative operation	Baseline
User interface	Basic task-selection process completed by a first-time user without assistance	Desirable

4 Conceptual Design

4.1 Overview

This section describes the core technical features and proposed implementation of the assistive backpack. The backpack will be composed of five key subsystems each interacting with each other to deliver an intuitive user experience that reduces the organisational stresses of daily tasks. The architecture and preliminary calculations include some room for expansion.

The proposed system will use an Arduino as the central control unit. User inputs come from the task-selection interface and RFID scanner. The backpack will com-

municate via the LCD, LEDs and buzzer, the system architecture is shown below in Figure 1.

4.2 Design Rationale

A morphological design approach was used to compare possible solutions for the major system functions. The selected solutions prioritise feasibility, ease of use and integration into the backpack.

Table 2: Morphological design selections.

Function	Potential approaches	Choice / Reason
Task selection	Button selector, Rotary encoder, Joystick, Voice Activated, Clock based, Web-app	Button selector, Low cost, visually intuitive and easily integrated into the backpack. More practical than voice, web-based or timing-based alternatives.
Output display	Liquid Crystal, LED, buzzer, LED Matrix, Web-app, Haptics	LED + LCD + Buzzer, Provides both visual and audible feedback. The LCD communicates more information than LEDs alone while remaining simpler than a web interface. Slightly more power draw.
Item tracking	RFID, QR-code, Barcode, Weight-based	RFID Module, The RC522 is a low-cost Arduino-compatible design that is simple to implement and does not require line-of-sight, item orientation or additional imaging/camera hardware. Passive tags require no onboard power.
Power	Battery, Solar, Mains power, Nuclear	Battery, Provides portability and does not rely on mains power or environmental conditions for operation.

4.3 Subsystem Breakdown and Rationale

The function and proposed implementation of each subsystem are summarised below.

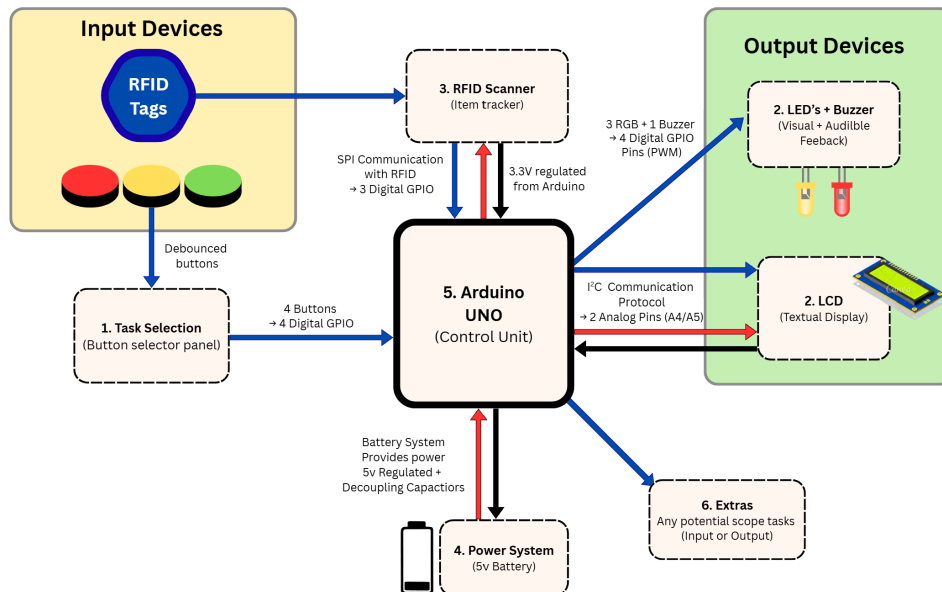
**Figure 1:** Proposed system architecture and information flow.

Table 3: Subsystem functions and proposed implementations.

Subsystem	Function	Inputs	Outputs	Implementation
1 Task Selection	Allows users to select daily task presets and start or stop task sessions.	User button presses	Button commands to Arduino	Four buttons (<i>Next</i> , <i>Previous</i> , <i>Start</i> , <i>Stop</i>) provide menu navigation and task control. Inputs are hardware debounced before being processed by the Arduino.
2 Output Display	Provides visual and audible feedback for task selection, item tracking and session status.	Commands and status information from Arduino	Textual, visual and audible feedback to user	A 16×2 character LCD displays task and item information, while an RGB LED and buzzer provide additional visual and audible feedback.
3 Item Scanner	Detects required items and assists with backpack inventory tracking.	RFID tags attached to items; Arduino control signals	Identified item ID to Arduino	An RC522 RFID reader detects passive RFID tags without requiring line-of-sight. The identified tag is communicated to the Arduino using SPI.
4 Power System	Supplies regulated electrical power to the backpack electronics.	Rechargeable battery	Regulated power rails for system components	A rechargeable battery provides portable power. Voltage regulation will provide the required supply voltages for individual components, with battery-voltage monitoring considered for a future low-battery warning.
5 Arduino	Acts as the central logic controller and coordinates system operation.	Button inputs, RFID data, and potential future inputs	LCD commands, RGB LED control, buzzer signals and potential future outputs	The Arduino executes a finite-state machine (FSM) managing states such as <i>Task Selection</i> , <i>Item Acquisition</i> and <i>Task Active</i> . Communication with peripherals uses digital I/O, SPI and I ² C interfaces as appropriate.

4.4 Preliminary Power Calculations

A preliminary power budget was calculated to establish the approximate battery capacity required by the proposed electronics (with an extra 20mA budgeted for stretch goal components). The maximum estimated continuous current is shown in 4.

Table 4: Preliminary maximum continuous current budget.

Component	Current
Arduino[7], [8], [9]	50 mA
RGB LED[10]	60 mA
16×2 LCD[11], [12]	50 mA
RC522 RFID[13]	20 mA
Buzzer[14]	20 mA
Stretch Goals / Extra	20 mA
Total	220 mA

The battery capacity for a target operating time of 10 hr, while limiting the nominal battery utilisation to 80% (for any overheads), is given by

$$C_{\text{battery}} = \frac{I_{\text{total}}t}{0.8} = \frac{(220)(10)}{0.8} \approx 2750 \text{ mAh.} \quad (1)$$

The goal is to also operate most subsystems at 5 V which avoids unnecessary energy losses. The RFID Module however runs at 3.3 V; possibly powered by the Arduino’s onboard 3.3 V pin. Due to the linear regulator this voltage step down is only about 66% efficient, a 5.5V → 3.3V buck convertor could be used in its place to reduce thermal losses. Combining this information, a preliminary battery target of approximately 2750 mAh running at a nominal 5 V is appropriate. The power architecture is not final; this model gives an overestimate for power consumption of the finished product. The battery capacity and voltage regulator design will depend on integrated testing and final voltage requirements.

4.5 Device Interface Considerations

The proposed electrical interfaces were selected to minimise Arduino pin usage while retaining sufficient flexibility for future expansion. The multitude of sensors/devices used in this project would benefit strongly from a shared communication bus. Multiple I²C devices can share the same bus using addressing, freeing up many pins on the Arduino.

Table 5: Preliminary electrical interfaces.

Component	Interface	Processing
Buttons	4 Digital (4 Pin)	Software debounce or capacitor + Schmitt trigger before FSM processing.
Buzzer	Digital / PWM (1 Pin)	Arduino output provides audible notifications and user feedback.
RC522	SPI (4 pin)	Shared SPI bus requiring approximately four interface signals.
LCD	Parallel or I ² C (2 pin)	I ² C module preferred to reduce pin usage; parallel operation requires 10 or more digital pins.
RGB LED	PWM (3 RGB pins)	Arduino PWM outputs control the individual colour channels.
Stretch Goals (Optional)	I ² C (2 Pins) / SPI (4 pin)	Multiple communication lines depending on type.

These preliminary communication designs suggest a total pin count of:

$$4_{\text{Buttons}} + 1_{\text{Buzzer}} + 5_{\text{RC522}} + 2_{\text{LCD}} + 3_{\text{LED}} + 2_{\text{HW123}} = 17 \text{ I/O pins.}$$

This fits within the 20 pin limit of the Arduino Uno, given that the I²C protocol can be implemented for the LCD screen. By sharing I²C or SPI bus lines the pin usage can be minimised, leaving some room for future expansion.

5 Project Management

5.1 Timeline

The Gantt Chart maps the major work tasks to semester weeks and the five ESI milestones, with the dependencies shown as link arrows. Component ordering begins as soon as the design is locked in Week 4, since the battery, charger and buck converter are not supplied in-kit and must arrive before build work starts in Week 6.

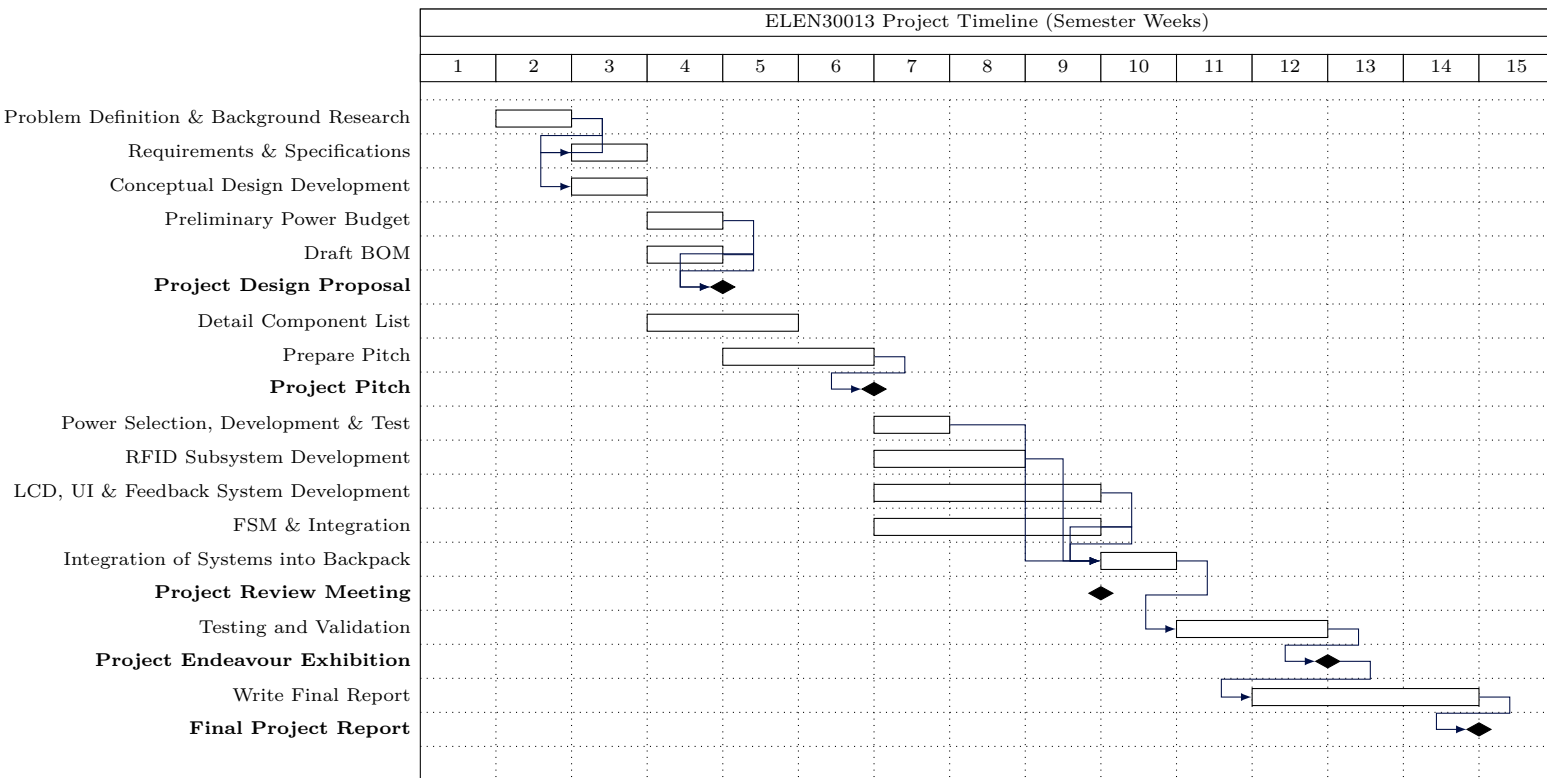


Figure 2: Project Gantt chart with task dependencies mapped to semester weeks and ESI milestones.

5.2 Roles and Responsibilities

Each work area has exactly one Responsible owner with final decision authority, while other members remain Consulted or Informed, following the RACI structure defined in the team charter (Table 6). The team meets weekly (1 hour) with a short asynchronous mid-week check-in, plus additional sessions during and after workshop sessions. Decisions and action items are recorded on the meeting minutes document found on the Microsoft teams.

Table 6: Responsibility RACI Matrix.

Task	Emerson	Callum	Oliver
Requirements & documentation	R	C	C
RFID / item-detection subsystem	R	C	I
LCD, UI & feedback subsystem	I	R	C
Firmware (FSM & integration)	I	I	R
Power system & battery	C	I	R
Enclosure & backpack build	R	C	I
Testing & validation	C	R	C
Final report & presentation	C	R	C

R = Responsible (final decision), C = Contributes (hands-on work),
I = Informed

5.3 Preliminary Bill of Materials

Currently all preliminary materials and components are part of the Super Learning Lab Kit For Arduino or belong to the students [15]. We will extensively be using already available materials to reduce our impact on the environment. If future purchases are required we will update the BOM accordingly.

Table 7: Preliminary BOM (AUD 100 budget cap).

Item	Quantity	Cost (\$)
1. Arduino Uno	1	In-kind
2. RGB LED	1-2	In-kind
3. 16x2 I ² C LCD	1	In-kind
4. RC255 Module	1	In-kind
5. RFID Tags	≥ 3	3 In-kind
6. Buzzer	1	In-kind
7. Push Buttons	4	In-kind
8. Resistors	tbc	In-kind
9. Decoupling Caps	tbc	In-kind
10. 2750mAh 5V or 7.4V Battery	1	In-kind
Total	15	\$0

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Appendix: Contribution Statement

A few sentences from each team member describing their contribution to the proposal.

- Emerson Gammaldi: Wrote sections 1 and 2 (Project Overview and Background), including main project discussion and project scoping.
- Callum Storrar: Wrote Section 3 (Requirements and Specifications) and Section 5 (Project Management). This involved making the specifications and verification targets, Gantt Chart, RACI matrix and working with Oliver to complete the BOM table.
- Oliver McKinlay : Wrote section 4 (Conceptual Design), including making the subsystem diagram, doing prelim. power calculations and interfacing protocols. Was also part of background and overview discussions with Emerson in team meeting.

Declaration

We declare this report is our own work and has been prepared and compiled in accordance with the university's academic integrity requirements. Generative AI tools were used in a limited manner during the writing of this report for clarity and formatting. AI tools were not used to replace our own analysis, calculations, or design decisions. All external sources have been appropriately acknowledged.